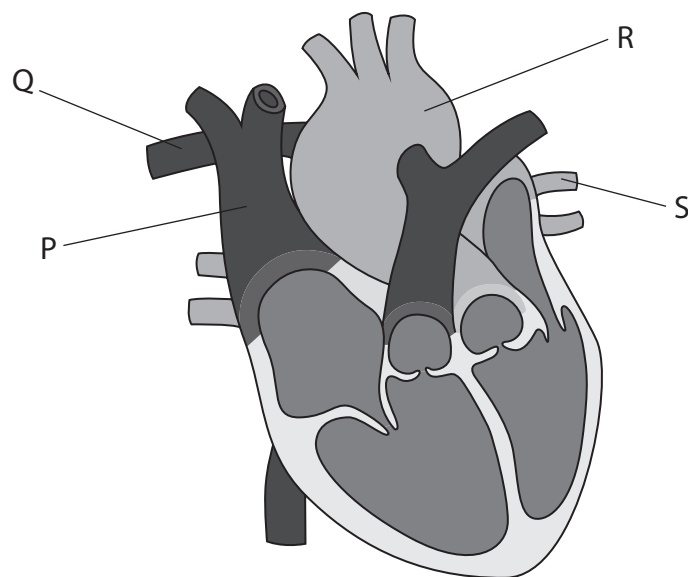


4 Many animals have a heart and circulation.

(a) The diagram shows a human heart and its major blood vessels, labelled P, Q, R and S.



(i) Which pair of blood vessels transports blood to and from the lungs?

(1)

- ☐ A P and Q
- ☐ B Q and S
- ☐ C R and P
- ☐ D S and R

(ii) Which blood vessels have valves along their length?

(1)

- ☐ A P
- ☐ B P and R
- ☐ C Q
- ☐ D Q and R

(iii) Which blood vessel does the coronary artery branch from?

(1)

- ☐ A P
- ☐ B Q
- ☐ C R
- ☐ D S



P 7 5 7 7 1 A 0 9 2 8

(b) Capillaries enable the movement of molecules between the blood and cells.

The distance the molecules diffuse between the blood and cells affects the rate of diffusion. This is the diffusion distance.

(i) Name **two** other factors that affect the rate of diffusion.

(1)

1

2

(ii) Name the law of diffusion that can be used to calculate the rate of diffusion.

(1)

.....

(c) The table shows the time taken for a molecule to diffuse over different diffusion distances.

Diffusion distance	Diffusion distance / μm	Time taken to diffuse the distance	Time taken to diffuse the distance / milliseconds
1 μm	1	0.5 milliseconds	0.5
10 μm	10	50.0 milliseconds	50.0
100 μm	100	5.0 seconds	
1 mm		8.3 minutes	

(i) Complete the table to show the missing values.

(2)



- (ii) Explain why it is important for body cells to be close to capillaries to meet the demands of respiration.

Use the information in the question and the table to support your answer.

(3)

(Total for Question 4 = 10 marks)

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